Cathepsin A-Related Arteriopathy with Strokes and Leukoencephalopathy (CARASAL)

Disease: Cathepsin A-Related Arteriopathy with Strokes and Leukoencephalopathy (CARASAL) **MONDO ID:** MONDO:0035551 **Causal gene: CTSA** (cathepsin A / protective protein, PPCA; HGNC:2515), chromosome 20q13.12 **Category:** Mendelian, autosomal dominant

Summary

CARASAL is an ultra-rare, adult-onset, autosomal dominant hereditary cerebral small vessel disease (cSVD) caused by a single recurrent heterozygous missense variant, c.973C>T (p.Arg325Cys), in the *CTSA* gene on chromosome 20q13.12. The causal gene is **CTSA**, which encodes cathepsin A (also called protective protein/cathepsin A, PPCA), a lysosomal serine carboxypeptidase. It is essential to name the gene at the outset because *CTSA* is also the gene behind **galactosialidosis**, a completely different, autosomal *recessive* lysosomal storage disease. CARASAL and galactosialidosis are allelic disorders of the same gene but differ fundamentally in inheritance mode, mechanism, and clinical presentation: CARASAL arises from a dominant-acting missense allele that dysregulates endothelin-1 metabolism, whereas galactosialidosis arises from biallelic loss of cathepsin A's chaperone/protective function toward β-galactosidase and neuraminidase-1.

Mechanistically, cathepsin A is the major soluble endogenous enzyme that degrades the potent vasoconstrictor **endothelin-1 (ET-1)**. The p.Arg325Cys variant removes a buried, salt-bridged arginine residue from the enzyme core, predicted to destabilize the protein fold. The resulting cathepsin A dysfunction leads to accumulation of ET-1, which was demonstrated directly in patient brains, where striking ET-1 immunoreactivity appeared in white-matter astrocytes and correlated with increased numbers of arrested premyelinating oligodendrocyte progenitors. This drives two convergent pathological streams: (1) ET-1–mediated arteriolar vasoconstriction producing therapy-resistant hypertension and small-vessel arteriopathy, and (2) disproportionately severe leukoencephalopathy with arrested oligodendrocyte maturation.

The clinical picture is dominated by ischemic and hemorrhagic strokes, slow late-onset cognitive decline, therapy-resistant hypertension, migraine, and brainstem/cranial-nerve features (hearing loss, tinnitus, non-positional vertigo, facial pain, dysphagia, dysarthria).

To date, approximately 19–21 patients have been reported across Dutch, Italian/Brazilian, and Chinese pedigrees, all carrying the identical p.Arg325Cys allele — making CARASAL both allelically homogeneous and extraordinarily rare. There is no disease-modifying therapy; management is symptomatic. Because the mechanism converges on the endothelin pathway, ET-1 receptor antagonism is a rational (but untested) therapeutic hypothesis. This report synthesizes seven confirmed findings and 37 reviewed papers into a comprehensive disease-characteristics entry spanning etiology, phenotypes, molecular genetics, mechanism, anatomy, temporal course, epidemiology, diagnostics, prognosis, treatment, prevention, comparative biology, and model organisms.

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1. Disease Information

Overview. CARASAL is a monogenic, autosomal dominant cerebral small vessel disease (cSVD) characterized by adult-onset strokes (both ischemic and hemorrhagic), progressive leukoencephalopathy visible on MRI that is disproportionately severe relative to clinical status, slowly progressive cognitive decline, and therapy-resistant hypertension. It was first defined as a distinct clinical-genetic entity in 2016 following whole-exome sequencing of two Dutch families [PMID: 27664989]. The disease sits within the family of hereditary cSVDs alongside CADASIL (NOTCH3), CARASIL/HTRA1-related arteriopathy, COL4A1/2-associated microangiopathy, RVCL-S, and Fabry disease (GLA) [PMID: 42771568].

Key identifiers.

Resource	Identifier
MONDO	MONDO:0035551
Gene	CTSA (HGNC:2515); OMIM gene *613111*
Locus	Chromosome 20q13.12
Causal variant	NM_000308 c.973C>T, p.(Arg325Cys)
Disease type	Mendelian, autosomal dominant

Synonyms / alternative names. "Cathepsin-A-related arteriopathy with strokes and leukoencephalopathy"; the acronym **CARASAL**. The disease is sometimes grouped descriptively under "hereditary cerebral small vessel disease" and "monogenic vascular dementia." No widely used OMIM phenotype number, ICD-10/11, or MeSH term was recorded in the source material for CARASAL specifically; it is an ultra-rare, recently described entity.

Information source. The knowledge base here is derived from **aggregated disease-level resources and primary case reports / small cohort studies** (not population EHR). Evidence derives from a limited number of pedigrees and individual case descriptions, plus mechanistic studies in human tissue, in vitro systems, and mouse genetics.

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2. Etiology

Primary cause — genetic. CARASAL is caused by a **heterozygous founder missense variant in *CTSA*: c.973C>T, p.Arg325Cys** (Finding F001). Whole-exome sequencing of two Dutch families identified a single shared variant segregating with disease; haplotype analysis revealed a shared 1,145-kb interval on chromosome 20q13.12, indicating descent from a common ancestor (founder effect) [PMID: 27664989]. The same variant was subsequently confirmed in unrelated patients of Chinese and Italian/Brazilian ancestry [PMID: 35904593].

> *"we identified one variant shared by both families and segregating with the disease: c.973C>T in CTSA. Haplotype analysis revealed a shared 1,145-kb interval encompassing the CTSA variant on chromosome 20q13.12, suggesting a common ancestor"* — PMID: 27664989

Genetic risk factors. The disease is monogenic; the p.Arg325Cys allele is both necessary and (with age-dependent penetrance) essentially sufficient. No susceptibility loci or modifier genes have been established. One candidate biological modifier is **SCPEP1** (serine carboxypeptidase 1), a homolog that redundantly degrades ET-1 — in mice, only combined Ctsa/Scpep1 deficiency produces a strong blood-pressure phenotype [PMID: 24586188], raising the untested hypothesis that human SCPEP1 activity could modulate CARASAL severity.

Environmental risk factors. None established as causal. Standard vascular risk factors (hypertension — which is itself a disease feature — smoking, dyslipidemia) are plausible aggravators of small-vessel injury but have not been formally quantified in CARASAL. Age is the dominant determinant of clinical expression (adult/late onset).

Protective factors. No genetic or environmental protective factors have been identified. The redundant ET-1–degrading enzyme SCPEP1 is a theoretical protective modifier (see above).

Gene–environment interactions. Mouse data suggest a **salt–gene interaction**: catalytically inactive cathepsin A combined with Scpep1 loss produced significantly elevated blood pressure specifically *after a high-salt diet and systemic ET-1 injection* [PMID: 24586188]. By extrapolation, dietary salt and conditions that raise ET-1 could interact with impaired cathepsin A function to worsen hypertension in CARASAL (inferred, not demonstrated in patients).

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3. Phenotypes

CARASAL is dominated by central nervous system manifestations with onset in adulthood. Frequencies below are qualitative/semiquantitative given the very small reported cohorts (~19–21 patients).

Phenotype	Type	Onset / course	Frequency	Suggested HPO
Ischemic stroke	Clinical sign	Adult; episodic/ recurrent	Core feature	HP:0002140 (Ischemic stroke)
Intracerebral/ hemorrhagic stroke	Clinical sign	Adult; can be fatal	Reported	HP:0001342 (Cerebral hemorrhage)
Leukoencephalopathy (white-matter hyperintensities)	Imaging/lab abnormality	Adult; progressive; disproportionately severe	Core; ~universal	HP:0002352 (Leukoencephalopathy)
Therapy-resistant hypertension	Clinical sign	Adult; chronic	Core feature	HP:0000822 (Hypertension)
Cognitive decline / vascular dementia	Behavioral/ cognitive	Slow and late; progressive	Common	HP:0001268 (Mental deterioration)
Migraine	Symptom	Adult	Reported	HP:0002076 (Migraine)
Non-positional vertigo	Symptom	Adult	Reported	HP:0002321 (Vertigo)
Tinnitus	Symptom	Adult	Reported	HP:0000360 (Tinnitus)
Hearing loss	Clinical sign	Adult	Reported	HP:0000365 (Hearing impairment)
Facial pain	Symptom	Adult	Reported	HP:0030833 (Facial pain)
Dysphagia	Clinical sign	Adult	Reported	HP:0002015 (Dysphagia)
Dysarthria	Clinical sign	Adult	Reported	HP:0001260 (Dysarthria)
Disequilibrium / gait imbalance	Clinical sign	Adult	Reported	HP:0002141 (Gait imbalance)
Behavioral disinhibition	Behavioral	Adult	Reported	HP:0000734 (Disinhibition)
	Behavioral	Adult	Reported	

Phenotype	Type	Onset / course	Frequency	Suggested HPO
REM-sleep behavior disorder; depression				HP:0002360; HP:0000716 (Depression)
Enophthalmos / "orbital check-mark sign"	Physical/ imaging	Adult	Reported (2026 cohort)	HP:0000490 (Enophthalmos)

Clinical description. In the founding two families the clinical picture was *"dominated by ischemic and hemorrhagic strokes, slow and late cognitive deterioration, and therapy-resistant hypertension"* [PMID: 27664989] (Finding F003). A systematic review lists migraine, stroke with central facial palsy, facial pain, non-positional vertigo, cognitive dysfunction with impaired concentration and behavioral disinhibition, REM-sleep behavioral disorder, and depression [PMID: 31177426].

> *"These include migraine, stroke with central facial palsy, facial pain, non-positional vertigo, cognitive dysfunction with impaired concentration and behavioral disinhibition, REM-sleep behavioral disorder, and depression"* — PMID: 31177426

Severity and progression. Variable but progressive; the hallmark is a **disproportion** between very severe imaging burden and comparatively milder clinical disease. A 2026 cohort (21 patients, 3 pedigrees) quantified a higher normalized white-matter-hyperintensity volume than age/sex-matched CADASIL (mean difference 5.0% of intracranial volume, 95% CI 2.0–8.0) [PMID: 41908989] (Finding F003).

> *"Total nWMHv was higher in patients with CARASAL than in those with CADASIL (mean difference: 5.0% of intracranial volume, 95% CI [2.0-8.0]"* — PMID: 41908989

Quality of life. Not formally measured with standardized instruments (EQ-5D/SF-36) in the reported literature. By analogy to other cSVDs, recurrent stroke, progressive cognitive impairment, and gait/balance problems would be expected to produce substantial disability and QoL reduction.

4. Genetic / Molecular Information

Causal gene. *CTSA* (cathepsin A / protective protein PPCA), chromosome 20q13.12; HGNC:2515. The protein is a lysosomal serine carboxypeptidase (EC 3.4.16.5) of the α/β-hydrolase fold family, structurally homologous to yeast carboxypeptidase Y [PMID: 11212324].

Pathogenic variant (CARASAL).

Attribute	Value
cDNA	NM_000308 c.973C>T
Protein	p.(Arg325Cys)
Codon change	CGC → TGC (Arg → Cys)
Variant type	Missense (single-nucleotide substitution)
Classification	Pathogenic
Origin	Germline
Zygosity in disease	Heterozygous (single allele)
Functional consequence	Protein destabilization / folding defect (predicted dominant-acting)

Allelic homogeneity. p.Arg325Cys is the **sole reported CARASAL allele** (Finding F006). A systematic literature review returns no CARASAL family carrying any other *CTSA* variant; all reported pedigrees (Dutch, Italian/Brazilian, Chinese) carry this same recurrent allele [PMID: 27664989; 35904593; 31177426; 41908989].

> *"Both patients carried the same pathogenic p.Arg325Cys mutation in CTSA"* — PMID: 35904593

Structural consequence (in silico, Findings F005 & F007). Mapping the variant onto the human cathepsin A/PPCA crystal structure (PDB 1IVY) shows that the mutated arginine (UniProt P10619 Arg326 = PDB residue 298; reference sequences differ by one N-terminal residue, and the Arg→Cys codon change confirms the mapping) is **partially buried** (156 heavy atoms within 10 Å of its guanidinium carbon) and participates in a **salt-bridge / hydrogen-bond network** with acidic side chains Glu264 (3.47 Å), Asp300 (3.69 Å), and Asp332 (5.83 Å). Removing this buried charged residue is predicted to destabilize the fold. The nearest free

cysteine thiol (Cys334 Sγ) lies ~9 Å away — too distant for a direct non-native disulfide. Critically, the variant sits **18.3 Å from the catalytic Ser150 hydroxyl** and **17.1 Å from the opposing subunit** (homodimer interface), i.e., it is neither in the catalytic pocket nor at the dimer interface. This favors a **folding/stability mechanism** over direct catalytic-site disruption.

This interpretation is reinforced by the published structural analysis of galactosialidosis mutations, which found that pathogenic *CTSA* missense residues act by perturbing structural integrity rather than directly hitting the active site:

> *"analysis of the 11 mutations revealed a substantive correlation between the effect of the amino acid substitution on the integrity of protein structure and the general severity of the clinical phenotype"* — PMID: 9435242
>
> *"none of the mutations occurred in the active site or at the protein surface, which would have disrupted the catalytic activity or protective function"* — PMID: 9435242

Conservation. The disease arginine is evolutionarily conserved (Finding F006). The local motif is human SGDKVR(M)DPPCT vs mouse SGDKVR(L)DPPCT — the disease arginine and the adjacent disulfide cysteine are invariant, with only a single conservative flanking M327→L difference.

Modifier genes. None confirmed. **SCPEP1** is a strong biological candidate based on its redundant ET-1–degrading role in mice [PMID: 24586188].

Epigenetic information / chromosomal abnormalities. No CARASAL-specific DNA-methylation, histone-modification, or structural/chromosomal abnormalities have been reported. CARASAL is a point-mutation disorder; no copy-number or cytogenetic changes are implicated.

Contrast with galactosialidosis. Recessive *CTSA* variants (e.g., frameshifts c.517delTT, intronic splicing defects; founder p.Pro203Thr in the Bahraini population) cause galactosialidosis — combined deficiency of neuraminidase-1 and β-galactosidase — via loss of the protein's protective/chaperone function [PMID: 9603439; 39981487; 11212324]. CARASAL's single heterozygous missense allele does not produce galactosialidosis, underscoring a distinct, dominant-acting mechanism.

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5. Environmental Information

- **Environmental factors:** None established as causal. Dietary salt is a plausible aggravator of the hypertensive component by analogy to mouse data [PMID: 24586188] (inferred).
- **Lifestyle factors:** Conventional vascular risk factors (smoking, diet, physical inactivity) are generically relevant to small-vessel disease but not specifically studied in CARASAL.
- **Infectious agents:** Not applicable. CARASAL is a non-infectious monogenic disorder.

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6. Mechanism / Pathophysiology

Ordered causal chain

1. **Germline heterozygous *CTSA* c.973C>T (p.Arg325Cys)** substitutes a buried, salt-bridged arginine with cysteine → **destabilizes the cathepsin A fold** (in silico; the residue is 18 Å from the active site and forms salt bridges to Glu264/Asp300/Asp332) [Findings F005, F007; P 9435242].
2. Protein destabilization **leads to dysfunction of cathepsin A's serine-carboxypeptidase activity** toward endothelin-1 (inferred from structural data and from the enzyme's established role as the principal ET-1–degrading enzyme).
3. Impaired degradation **results in accumulation of endothelin-1 (ET-1)** — cathepsin A/PPCA is *"a major soluble endogenous endothelin degradation enzyme in human tissues"* [PMID: 7822272] (Finding F002).
4. ET-1 accumulation **branches** into two downstream streams:
5. **Branch A (vascular):** ET-1 (a potent vasoconstrictor) → heightened arteriolar vasoconstriction and increased ET-1–dependent phosphorylation of myosin light chain 2 in vascular smooth muscle → **therapy-resistant hypertension** and small-vessel arteriopathy (severe arteriolar "atherosclerosis") [PMID: 24586188; 27664989].
6. **Branch B (glial/white matter):** ET-1 accumulates in **white-matter astrocytes** → correlates with **increased premyelinating oligodendrocyte progenitors** (arrested oligodendrocyte maturation) → impaired myelination/white-matter integrity [PMID: 27664989].

7. Both branches converge → **chronic small-vessel ischemia + primary white-matter/ oligodendrocyte dysfunction → disproportionately severe leukoencephalopathy** with secondary ischemic and hemorrhagic strokes [PMID: 27664989; 31177426].
8. Cumulative vascular and white-matter injury → **progressive cognitive decline / vascular dementia**, brainstem and cranial-nerve features (hearing loss, tinnitus, vertigo, dysphagia, dysarthria) [PMID: 31177426; 36324419].

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CTSA p.Arg325Cys (heterozygous)
         |
  destabilizes cathepsin A fold  (18 A from active site; salt-bridge loss)
         |
  impaired ET-1 degradation
         |
  ENDOTHELIN-1 ACCUMULATION
       /                          \
Branch A (vascular)          Branch B (glial)
arteriolar vasoconstriction   ET-1 in WM astrocytes
^ MLC2 phosphorylation        arrested OPC maturation
therapy-resistant HTN         (^ premyelinating OPCs)
small-vessel arteriopathy     impaired myelination
        \                          /
         \________________________/
                    |
  small-vessel ischemia + primary WM injury
                    |
  DISPROPORTIONATELY SEVERE LEUKOENCEPHALOPATHY
  ischemic + hemorrhagic strokes
  progressive cognitive decline; brainstem/cranial-nerve signs
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Detail by category

- **Molecular pathways.** Endothelin signaling (ET-1 → ETA/ETB receptors → Gq → PLC → ↑intracellular Ca^{2+} → myosin light chain phosphorylation → vasoconstriction). Mouse mesenteric arteries lacking cathepsin A showed *"increased ET-1-dependent phosphorylation of myosin light chain 2"* [PMID: 24586188].
- **Cellular processes.** Vascular smooth-muscle contraction/arteriopathy; astrocyte ET-1 handling; **arrested oligodendrocyte precursor differentiation** (a developmental/ maturation block rather than overt demyelination).
- **Protein dysfunction.** Loss-of-stability / folding defect of cathepsin A (Findings F005/F007). Because CARASAL is heterozygous and dominant-acting while galactosialidosis is recessive, the CARASAL allele likely acts by a mechanism other than simple haploinsufficiency of the

protective function — plausibly a dominant effect on ET-1 regulation or a subtle dominant-negative/altered-processing effect (the precise dominant mechanism remains to be demonstrated).

- **Metabolic / biochemical abnormalities.** Reduced carboxypeptidase-mediated proteolysis of ET-1; *"the circulating level of ET-1 is regulated through its proteolytic cleavage by secreted serine carboxypeptidase, cathepsin A"* [PMID: 24586188].
- **Immune involvement.** No primary autoimmune/immunodeficiency mechanism; chronic ischemic/gliotic injury may involve reactive astrogliosis and microglial activation (as in cSVD generally) but is not the initiating lesion.
- **Tissue damage mechanisms.** Ischemia from small-vessel vasoconstriction/arteriopathy plus intrinsic white-matter oligodendrocyte dysfunction; hemorrhage from fragile small vessels.
- **Molecular profiling.** The key human-tissue finding is immunohistochemical: striking ET-1 immunoreactivity in white-matter astrocytes with increased premyelinating oligodendrocytes [PMID: 27664989]. No large-scale transcriptomic/proteomic/metabolomic datasets specific to CARASAL were identified.

> *"One of the numerous cathepsin A functions is to degrade endothelin-1. In the patients, striking endothelin-1 immunoreactivity was found in white matter astrocytes, correlating with increased numbers of premyelinating oligodendrocyte progenitors"* — PMID: 27664989

Upstream vs downstream. Upstream: the *CTSA* mutation → protein destabilization → ET-1 accumulation. Downstream: vasoconstriction/hypertension, arrested myelination, ischemia, strokes, cognitive decline.

Suggested GO / CL terms. GO:0006508 (proteolysis); endothelin receptor signaling pathway; GO:0042310 (vasoconstriction); GO:0048709 (oligodendrocyte differentiation). Cell types — CL:0000128 (oligodendrocyte), CL:0002453 (oligodendrocyte precursor cell), CL:0000127 (astrocyte), CL:0000359 (vascular smooth muscle cell), CL:0000115 (endothelial cell).

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7. Anatomical Structures Affected

Organ / system level. - **Primary:** central nervous system — cerebral white matter and small cerebral arteries/arterioles (nervous + cardiovascular systems). - **Secondary:** cognitive networks, brainstem, inner ear/vestibular pathways. - **Body systems:** nervous (CNS), cardiovascular (systemic hypertension, small-vessel arteriopathy).

Specific anatomical sites (with UBERON suggestions). MRI shows *"extensive signal change in the deep cerebral white matter, anterior temporal lobes, thalami, internal and external capsules and brainstem"* [PMID: 35904593] (Finding F003).

Site	UBERON
Cerebral deep white matter	UBERON:0002316 (white matter)
Anterior temporal lobe	UBERON:0002771 (temporal lobe region)
Thalamus	UBERON:0001897
Internal capsule	UBERON:0022279
External capsule	UBERON:0022392
Brainstem / pons	UBERON:0000988 (pons)
Cerebral arteriole	UBERON:0001637 (artery) / arteriole
Orbit (enophthalmos / "orbital check-mark sign")	UBERON:0001697 (orbit)

Tissue / cell level. Nervous tissue (white matter) and vascular wall. Cell populations: **oligodendrocytes and oligodendrocyte precursor cells** (arrested maturation), **astrocytes** (ET-1 accumulation), **vascular smooth muscle cells** and **endothelial cells** of small vessels. CL terms: CL:0000128, CL:0002453, CL:0000127, CL:0000359, CL:0000115.

Subcellular level. Cathepsin A is a **lysosomal** enzyme (GO:0005764 lysosome); it is also secreted and acts extracellularly/in the circulation on ET-1 [PMID: 7822272; 24586188]. GO cellular-component terms: GO:0005764 (lysosome), GO:0005615 (extracellular space).

Lateralization. White-matter and small-vessel involvement is **bilateral**, typically symmetric/diffuse (characteristic of leukoencephalopathy).

8. Temporal Development

- **Onset:** Adult / late-onset. Imaging-based diagnosis in the founding cohort was applied to family members ≥40 years [PMID: 27664989]. Onset is **insidious/chronic** with superimposed **acute** episodes (strokes).
- **Progression:** Progressive, with episodic stroke events punctuating a chronically progressive leukoencephalopathy and cognitive decline. Cognitive deterioration is described as *"slow and late"* [PMID: 27664989].
- **Course pattern:** Chronic, lifelong, progressive with episodic strokes.
- **Disease stages:** Not formally staged; conceptually early (imaging changes, hypertension) → intermediate (recurrent strokes, early cognitive change) → advanced (vascular dementia, disability).
- **Remission:** None; no spontaneous or treatment-induced remission reported.
- **Critical periods:** Adulthood onward; the pre-symptomatic adult window (when imaging changes may precede clinical events) is the theoretical opportunity for intervention.

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9. Inheritance and Population

Epidemiology. Ultra-rare. Approximately **19 patients** reported as of the 2019 systematic review [PMID: 31177426]; a 2026 cohort described 21 patients across 3 pedigrees [PMID: 41908989]. No reliable prevalence/incidence estimates exist; CARASAL is likely underdiagnosed and expected to increase in recognition with broader genetic testing and advanced imaging [PMID: 31177426].

> *"So far, 19 patients have been reported. The frequency of CARASAL patients will most likely increase in the future, as CARASAL may be more frequently recognized"* — PMID: 31177426

Inheritance. Autosomal dominant, single heterozygous allele (Findings F001, F006).

Penetrance / expressivity. Appears high but age-dependent (adult onset); expressivity is variable across the reported phenotype spectrum. Formal penetrance estimates are not available given tiny cohorts.

Genetic anticipation / germline mosaicism. Not reported / not applicable (point mutation, not a repeat expansion).

Founder effect. Yes — a shared 1,145-kb haplotype on 20q13.12 in the Dutch families indicates a common ancestor [PMID: 27664989]. Remarkably, the identical p.Arg325Cys allele recurs in unrelated patients of Chinese and Italian/Brazilian ancestry [PMID: 35904593], consistent either with multiple independent mutational events at this CpG-containing codon (CGC→TGC is a classic C>T transition at a methylated CpG) or broader distribution.

Consanguinity. Not relevant (dominant disorder).

Carrier frequency. Not applicable in the recessive sense; the pathogenic dominant allele is extremely rare.

Population demographics. Reported in Dutch, Italian/Brazilian, and Chinese individuals. No established sex predilection. Age distribution: adults (≥40 years for most diagnosed cases).

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10. Diagnostics

Imaging (first-line). Brain **MRI** is central: diffuse, confluent, disproportionately severe white-matter hyperintensities involving deep cerebral white matter, **anterior temporal lobes**, thalami, internal/external capsules, and brainstem/pons [PMID: 35904593]. The 2026 cohort quantified a higher normalized WMH volume than CADASIL (mean difference 5.0% of ICV, 95% CI 2.0–8.0) and noted an **"orbital check-mark sign"/enophthalmos** as a potential supportive feature [PMID: 41908989]. Microbleeds/hemorrhage may be seen on susceptibility-weighted imaging.

Genetic testing (confirmatory, definitive). Because CARASAL is allelically homogeneous, **targeted single-variant / single-gene *CTSA* testing** for c.973C>T (p.Arg325Cys) is the most efficient confirmatory approach. **Whole-exome sequencing (WES)** was the discovery method and remains valuable when the diagnosis is not clinically suspected [PMID: 27664989]; **whole-genome sequencing (WGS)** and gene panels for hereditary cSVD (which should include *CTSA* alongside NOTCH3, HTRA1, COL4A1/2, GLA, TREX1) are alternatives. Chromosomal microarray, karyotyping, FISH, mtDNA, and repeat-expansion testing are **not applicable**.

Laboratory / biomarkers. No validated blood/CSF biomarker. **Endothelin-1** is a mechanistically rational candidate circulating/tissue biomarker but is not an established clinical test. Cathepsin A enzyme-activity assays are used for galactosialidosis diagnosis but their utility in CARASAL (where the protective function may be partly preserved) is unproven.

Biopsy / pathology. Brain autopsy in the founding families showed a **primary microangiopathy with severe arteriolar changes and secondary leukoencephalopathy**, with leukoencephalopathy *disproportionately extensive relative to the vascular abnormalities*, and immunohistochemistry demonstrating ET-1 in white-matter astrocytes with increased premyelinating oligodendrocytes [PMID: 27664989; 31177426].

Clinical criteria / differential diagnosis. No formal diagnostic criteria. Key differentials are other hereditary cSVDs and monogenic vascular dementias:

Condition	Gene	Distinguishing features
CADASIL	NOTCH3	Anterior temporal/external capsule WMH; granular osmiophilic material; cysteine-altering NOTCH3 variants [PMID: 40145673]
CARASIL	HTRA1	Recessive; alopecia, spondylosis
COL4A1/2 microangiopathy	COL4A1/2	Hemorrhage-prone, ocular/renal features [PMID: 41005670]
Fabry disease	GLA	X-linked; angiokeratoma, renal/cardiac, acroparesthesia [PMID: 39978321]
MELAS	mtDNA m.3243A>G	Stroke-like episodes, lactic acidosis, maternal inheritance [PMID: 36324419]
Galactosialidosis	CTSA (recessive)	Coarse facies, cherry-red spot, visceral storage — distinct allelic disorder [PMID: 39981487]

CARASAL is distinguished by therapy-resistant hypertension, the disproportionately heavy white-matter burden, brainstem involvement, and the specific *CTSA* p.Arg325Cys genotype.

Screening. Cascade (predictive) genetic testing of at-risk adult relatives for the familial p.Arg325Cys allele is feasible. No population or newborn screening is indicated (ultra-rare, adult-onset, no preventive therapy).

11. Outcome / Prognosis

- **Survival / mortality.** No formal survival statistics. The disease is progressive and includes **potentially fatal intracerebral hemorrhage** [PMID: 41908989]. Life expectancy is likely reduced by recurrent stroke and vascular dementia, but quantitative data are lacking (too few cases) [PMID: 31177426].

> *"Currently, the outcome cannot be sufficiently assessed, as too few cases have been reported."* — PMID: 31177426

- **Morbidity / disability.** Substantial — recurrent strokes, progressive cognitive decline/vascular dementia, gait/balance impairment, dysarthria, dysphagia, and sensory (hearing/vestibular) deficits accumulate disability over time.
- **Complications.** Recurrent ischemic and hemorrhagic strokes; vascular dementia; consequences of therapy-resistant hypertension.
- **Recovery potential.** Limited; the course is progressive without disease-modifying therapy.
- **Prognostic factors.** White-matter lesion burden (quantifiable via normalized WMH volume), stroke frequency/subtype, and hypertension control are plausible prognostic indicators. No validated prognostic biomarkers.

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12. Treatment

No disease-modifying therapy exists. Management is symptomatic and supportive [PMID: 31177426].

- **Pharmacotherapy (supportive).**
- **Antihypertensives** for the therapy-resistant hypertension (often difficult to control) — NCIT: Antihypertensive Agent.
- **Antiplatelet agents and statins** are used broadly in cSVD/stroke secondary prevention (as in CADASIL cohorts) [PMID: 42949550]; benefit in CARASAL is unproven and hemorrhage risk must be weighed given the hemorrhagic-stroke tendency.
- **Mechanism-based / rational (experimental, untested).** Because the pathophysiology converges on **endothelin-1 excess**, **endothelin-receptor antagonists** (e.g., bosentan,

ambrisentan, macitentan; NCIT: Endothelin Receptor Antagonist) are a biologically rational but **entirely hypothetical** therapeutic avenue. No clinical trials in CARASAL have been reported.

- **Advanced therapeutics (gene/cell/RNA/targeted/immuno).** None developed for CARASAL. Note: AAV-mediated gene therapy and enzyme replacement are in preclinical/clinical development for the *recessive* allelic disorder **galactosialidosis** [PMID: 34901309; 33426146; 28603679], but these replace lost protective function and are **not applicable** to the dominant CARASAL mechanism.
- **Surgical / interventional.** None specific; standard acute-stroke care applies.
- **Supportive / rehabilitative.** Physical, occupational, and speech/swallowing therapy; cognitive support; management of migraine, vertigo, hearing loss; aggressive vascular risk-factor modification.
- **Pharmacogenomics / personalized medicine.** Genotype-directed care is limited to confirming the diagnosis; no CARASAL-specific pharmacogenomic guidance exists.

Treatment outcomes. No response-rate or adverse-event data specific to CARASAL therapy are available.

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13. Prevention

- **Primary prevention.** Not possible for mutation carriers (monogenic). Reproductive options (genetic counseling, prenatal/preimplantation genetic testing for the familial p.Arg325Cys allele) can prevent transmission.
- **Secondary prevention.** Cascade genetic testing and MRI surveillance of at-risk adult relatives for early identification; aggressive management of hypertension and vascular risk factors.
- **Tertiary prevention.** Stroke secondary-prevention measures and rehabilitation to limit complications and disability.
- **Immunization / public health / environmental interventions.** Not applicable (non-infectious, monogenic).
- **Counseling. Autosomal dominant genetic counseling** is central — 50% transmission risk to offspring, age-dependent expression, and discussion of predictive testing and reproductive options.

14. Other Species / Natural Disease

- **Taxonomy / orthologs.** The gene is conserved in mammals; mouse ortholog **Ctsa** (NCBI Gene; UniProt P16675). The disease arginine and adjacent disulfide cysteine are invariant between human and mouse (Finding F006).
- **Natural disease in other species.** No naturally occurring CARASAL-equivalent arteriopathy has been described in companion animals or wildlife in the reviewed literature. (Recessive *CTSA* loss-of-function models galactosialidosis in mice.)
- **Comparative biology / evolutionary conservation.** The cathepsin A → ET-1 regulatory axis is conserved: mouse genetics directly demonstrate cathepsin A control of blood pressure via ET-1 inactivation [PMID: 24586188], supporting conservation of the disease-relevant mechanism.
- **Transmission / zoonosis.** Not applicable.

15. Model Organisms

Available models. There is **no published mouse model that reproduces CARASAL** (the heterozygous p.Arg325Cys knock-in); the 2019 review explicitly flagged the need to generate animal models [PMID: 31177426]. Existing *Ctsa* mouse models recapitulate the **recessive** disorder (galactosialidosis), not CARASAL:

Model	Genotype	Phenotype modeled	Relevance to CARASAL
CathA S190A knock-in	Catalytically inactive cathepsin A (~10–15% residual ET-1 carboxypeptidase activity)	ET-1 handling; mild alone	Partial — isolates catalytic/ET-1 function [PMID: 24586188]
CathA/ Scpep1 double-deficient (DD)	Combined loss	↑BP after high-salt + ET-1; reduced blood ET-1 degradation; ↑MLC2 phosphorylation in mesenteric arteries	Mechanistic support for CARASAL hypertension [PMID: 24586188]
Ctsa-deficient (IVS6+1g/a) / Ctsa·Neu1	Splicing defect, combined deficiency	Galactosialidosis (myoclonus, visceral storage)	Models the recessive allelic disease, **not** CARASAL [PMID: 41917397; 35573044]

Phenotype recapitulation and limitations. The CathA/Scpep1 double-deficient mouse strongly supports the **vascular/hypertension branch** of CARASAL (Finding F004):

> *"in response to high-salt diet and systemic injections of ET-1 these animals showed significantly increased blood pressure as compared to wild type mice or those with single deficiencies of CathA or Scpep1"* — PMID: 24586188
>
> *"the circulating level of ET-1 is regulated through its proteolytic cleavage by secreted serine carboxypeptidase, cathepsin A"* — PMID: 24586188

However, **no model reproduces the dominant-acting p.Arg325Cys allele or the leukoencephalopathy / arrested-oligodendrocyte-maturation branch**, which is the defining CNS feature of CARASAL. A knock-in of the human-equivalent mutation (mouse Ctsa p.Arg→Cys) is the key missing resource.

Applications. Existing models are useful for studying cathepsin A's role in ET-1 metabolism and blood-pressure regulation; cellular systems (patient fibroblasts, iPSC-derived oligodendrocytes/astrocytes) would be needed to dissect the glial/white-matter mechanism.

Resources. MGI (mouse Ctsa), plus cell models (patient-derived fibroblasts; iPSC-derived neural cells) as future tools.

Mechanistic Model / Interpretation (synthesis)

CARASAL is best understood as a **dominant endothelin-1 dysregulation syndrome caused by a structurally destabilizing *CTSA* missense allele.** Three independent evidence streams converge:

1. **Genetics** pin the disease to a single recurrent heterozygous allele (p.Arg325Cys) that is both allelically homogeneous and evolutionarily conserved (F001, F006).
2. **Structural biology** predicts that this allele destabilizes the cathepsin A fold — the residue is buried, salt-bridged (Glu264/Asp300/Asp332), 18 Å from the active site, and away from the dimer interface — consistent with the established principle that pathogenic *CTSA* missense variants act via loss of structural integrity rather than direct catalytic-site hits (F005, F007).
3. **Functional biology** connects cathepsin A to endothelin-1: it is the principal soluble ET-1–degrading enzyme (F002), and mouse genetics prove that loss of this activity raises blood pressure and arterial ET-1 reactivity (F004). Patient tissue closes the loop by showing ET-1 accumulation in white-matter astrocytes with arrested oligodendrocyte maturation (F002/F003).

The result is a two-branch pathology — a **vascular branch** (ET-1–driven vasoconstriction → therapy-resistant hypertension + small-vessel arteriopathy) and a **glial branch** (astrocytic ET-1 → arrested oligodendrocyte maturation → white-matter failure) — that jointly explain the disease's signature: a leukoencephalopathy far more severe than the vascular lesions alone would predict.

Evidence Base

PMID	Title (abbrev.)	Role in this report
27664989	*CARASAL* (original description)	Identifies causal gene/variant (c.973C>T, p.Arg325Cys), founder haplotype, clinical triad, ET-1 in white-matter astrocytes with arrested OPCs (F001–F003)
35904593	*A rare cause of monogenic cSVD and stroke (CARASAL)*	Confirms recurrent p.Arg325Cys across ancestries; MRI distribution (F003, F006)
31177426	*Update on CARASAL*	Systematic review: 19 patients, phenotype spectrum, pathogenesis, need for models, outcome uncertainty
41908989	*Expanding the clinicoradiologic phenotype*	2026 cohort (21 patients); quantified WMH burden vs CADASIL; orbital sign (F003)
7822272	*Protective protein as endogenous endothelin degradation enzyme*	Establishes cathepsin A/PPCA as major ET-1–degrading enzyme (F002)
24586188	*SCPEP1 and Cathepsin A in vasoconstriction via ET-1 inactivation*	Mouse genetics: cathepsin A controls ET-1 and blood pressure; SCPEP1 redundancy (F004)
9435242	*Atomic model of human PPCA/ cathepsin A*	Structural basis; CTSA missense variants destabilize fold, lie away from active site (F005, F007)
11212324	*Cathepsin A/protective protein: multifunctional*	Enzyme biology, α/β-hydrolase fold, protective function
40145673	*Pathobiology of CADASIL*	Differential diagnosis (NOTCH3 cSVD)
42771568	*Hereditary cSVD: phenotype to management*	Positions CARASAL within hereditary cSVD; diagnostic clues

All quoted snippets above are verbatim from the cited abstracts.

Limitations and Knowledge Gaps

1. **Dominance mechanism unresolved.** Why a single heterozygous missense allele causes disease when galactosialidosis requires biallelic loss is not mechanistically proven (dominant-negative vs altered ET-1 regulation vs tissue-specific haploinsufficiency). The structural prediction of destabilization is in silico, not experimentally validated.
2. **No animal model of CARASAL.** All existing *Ctsa* mice model the recessive galactosialidosis phenotype; no knock-in reproduces p.Arg325Cys or the leukoencephalopathy.
3. **Tiny evidence base.** ~19–21 patients total; penetrance, expressivity, natural history, survival, and sex ratio are not reliably quantified.
4. **No validated biomarker or therapy.** ET-1 as a biomarker and ET-receptor antagonism as a therapy are mechanistically rational but untested.
5. **No omics datasets** (transcriptomic/proteomic/metabolomic) specific to CARASAL tissue beyond immunohistochemistry.
6. **Variant recurrence across ancestries** is unexplained — multiple independent CpG mutations vs shared ancestry remains unresolved.

Proposed Follow-up Experiments / Actions

1. **Generate a knock-in mouse** (Ctsa p.Arg→Cys heterozygous) to test whether the dominant allele recapitulates hypertension AND leukoencephalopathy, and to dissect dominant-negative vs loss-of-function contributions.
2. **Biochemically characterize p.Arg325Cys protein:** thermal stability, ET-1 carboxypeptidase kinetics, dimerization, and effect on NEU1/β-GAL complex — to confirm the predicted folding defect and test for a dominant-negative effect on the wild-type allele.
3. **Measure circulating and CSF ET-1** in patients vs controls to validate ET-1 as a diagnostic/prognostic biomarker.
4. **Model the glial branch in vitro** using patient iPSC-derived oligodendrocyte precursors and astrocytes to test whether ET-1 directly arrests oligodendrocyte maturation.
5. **Pilot endothelin-receptor antagonism** (preclinical first, in the knock-in model) as a mechanism-based therapeutic.

6. **Establish an international CARASAL registry** to accumulate natural-history, penetrance, and outcome data across the Dutch, Chinese, and Italian/Brazilian pedigrees and any newly identified carriers.
7. **Investigate SCPEP1** as a genetic/pharmacologic modifier of CARASAL severity.

Report compiled from 7 confirmed findings and 37 reviewed papers across 5 investigation iterations. Evidence types: human clinical (case reports/cohorts, autopsy, imaging), model organism (mouse genetics), in vitro (enzyme biology), and computational (structural modeling of PDB 1IVY).
